Appendix A2: conditional probability tables.

Contents

Network structure

Decision nodes

Controls the climate and fishing management scenarios, those are defined a priori depending on the conditions to be explored in the scenario analyses.

Extreme weather events (node: event)

The node *Event* is used to wrap all the extreme weather events we are considering.

Levels (and acronyms)

- **Storms** (sto). Strong wind events, with intensity above 24.5 m/s. Season: summer.
- **Gales** (gal), or moderately strong wind events, with intensity between 15 m/s and 24.5 m/s;
- **Extreme precipitation** (erf) – rainstorm (summer), or deviation from average precipitation regime;
- **Algal bloom** (abl): intense algal bloom;
- **Summer heatwaves** (hws): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Winter heatwaves** (hww): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Icing** (Ici): strong wind events associated to ice, which then forms icing pile up on gears and boats.

Fishing style (and acronyms)

The node *Fishing style* is used to wrap all the fishing styles and seasons.

Levels (and acronyms)

- **Recreational fisher** (rcf): a fishing style that operates year round. In winter, it primarily performs ice fishing and potentially can fish in open water from the shore; in summer fish with fishing rods both inland and in coastal area;
- **fishing guide** (fgu): a fishing style that operates year round. In winter, it primarily performs ice fishing and potentially can fish in open water from the shore; in summer fish with fishing rods both inland and in coastal areas;
- **Coastal small-scale fishers**, (ssf) a summer fishing style that use two kind of gillnets in coastal areas, and in some cases for salmon with fixed traps rigidly regulate. Gillnets types are herring gillnets (gns_spf) and freshwater gillnets (gns_fws). Traps are fpn_ana.
- **Pelagic trawlers** (trw), represented by pelagic trawlers active in winter and exclusively fishing for small pelagic with midwater trawl (otm);
- **Archipelago fishers** (arc), which is similar to an artisanal fishers but is by law prevented from selling their catches.

Fishing management

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **Business as Usual** (BAU): each fishing style adopts strategies and behaviour according to interviews results
- **Flexible management** (FLEX). Fishing style is allowed for more flexibility.

Stock status

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **status quo** (SQ): each fishing style adopts strategies and behaviour according to interviews results
- **improved stock status** (FLEX). Fishing style is allowed for more flexibility.
- **worsened stock status**

Chance nodes - Operational

Infrastructure

Parents: fishing_style.

Levels: no, some, extensive.

Description: availability of terrestrial infrastructures such as roads, ports and bridges. CPT is already filled with method M1 based on the answer to Q17 (infrastructure).

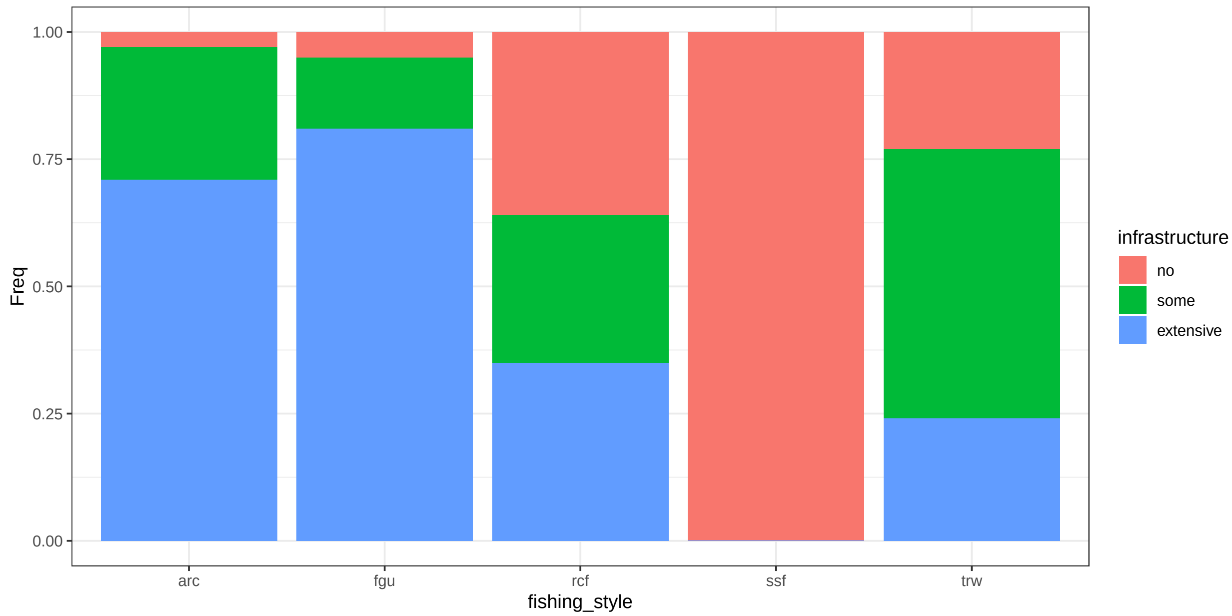

Figure B.1: conditional distribution for infrastructure .

Flexibility

Parents: management, infrastructure.

Levels: rigid, status_quo, flexible.

Description: choosing different flexibility possibility to change fishing strategy (both gear and target). This can be linked to the node "strategy".

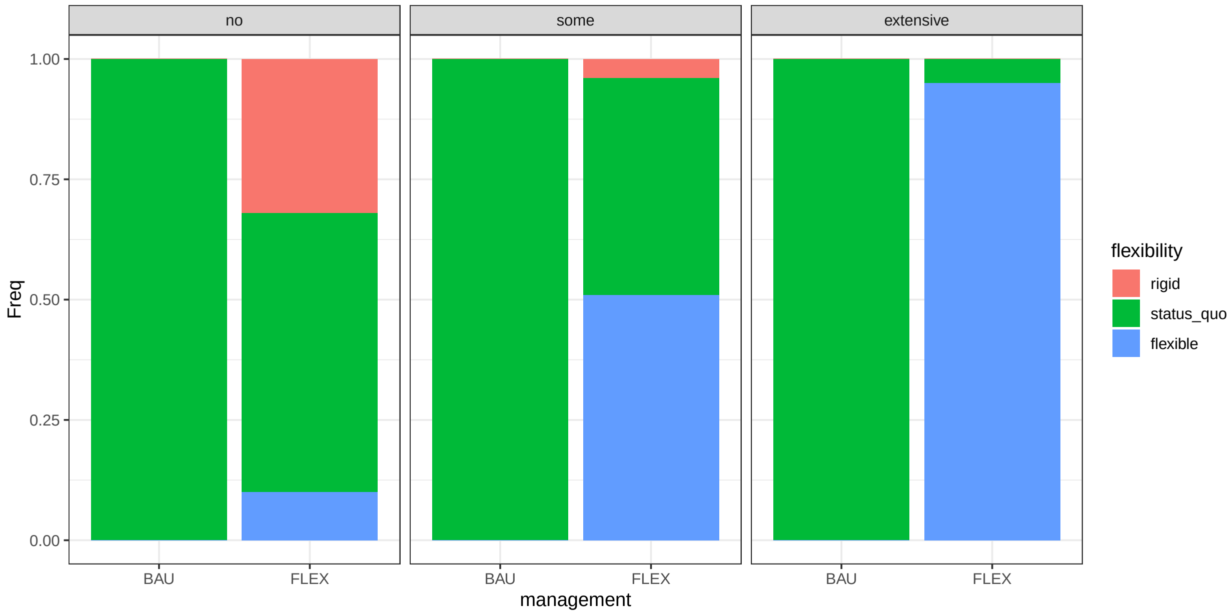

Figure B.2: conditional distribution for flexibility . Columns represents levels of the node infrastructure.

Strategy

Parents: fishing_style, event, flexibility.

Levels: cope, react, adapt, not_relevant.

Description: Fishers response is coded as Adapt (proactive planning); react (unplanned response); cope (accept passively) (Green et al., 2021). Cope is when fisher prefer to not go out fishing; React is when they apply small modification to their practice, such as fishing in deeper areas or in different time of the day; Adapt is when they take proactive response such as modify their strategy by changing fishing target and/or gear..

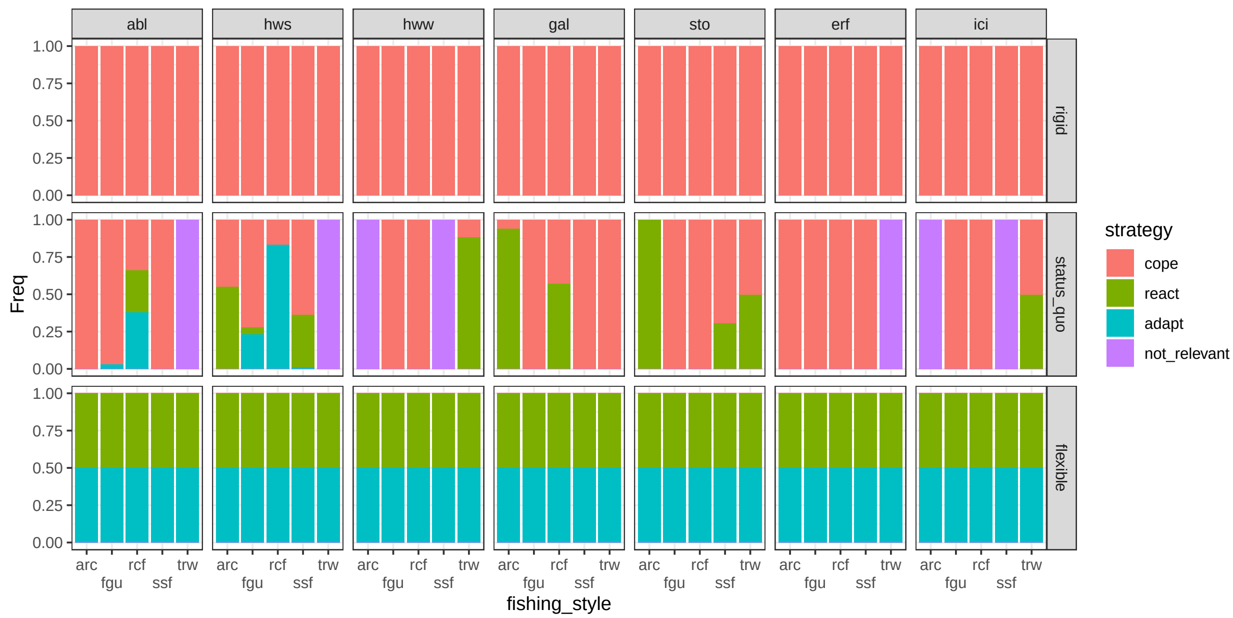

Figure B.3: conditional distribution for strategy . Columns represents levels of the node event; rows are levels of the node flexibility.

Additional mitigation

Parents: strategy, fishing_style, event.

Levels: no, travel_further, short_sets, other_technical.

Description: Describes the measures implemented in the case when the strategy is React. The different solutions work differently and have different cost: travel further implies fisherman change fishing location to seek a sheltered place or to find deeper waters; short sets mean that the fisher perform shorter sessions than average (i.e.: reduced soak time for gillnets); other technical includes strategies that are mentioned sporadically (i.e.: use of ice machines). The CPT is compiled manually for each combination of event and fisherman based on what has been mentioned throughout the questionnaire. In case only one strategy is mentioned, this is assigned probability = 1. When two strategies are mentioned, they are 50% probable unless one strategy is explicitly adopted more often than the other (and so on for three strategies)..

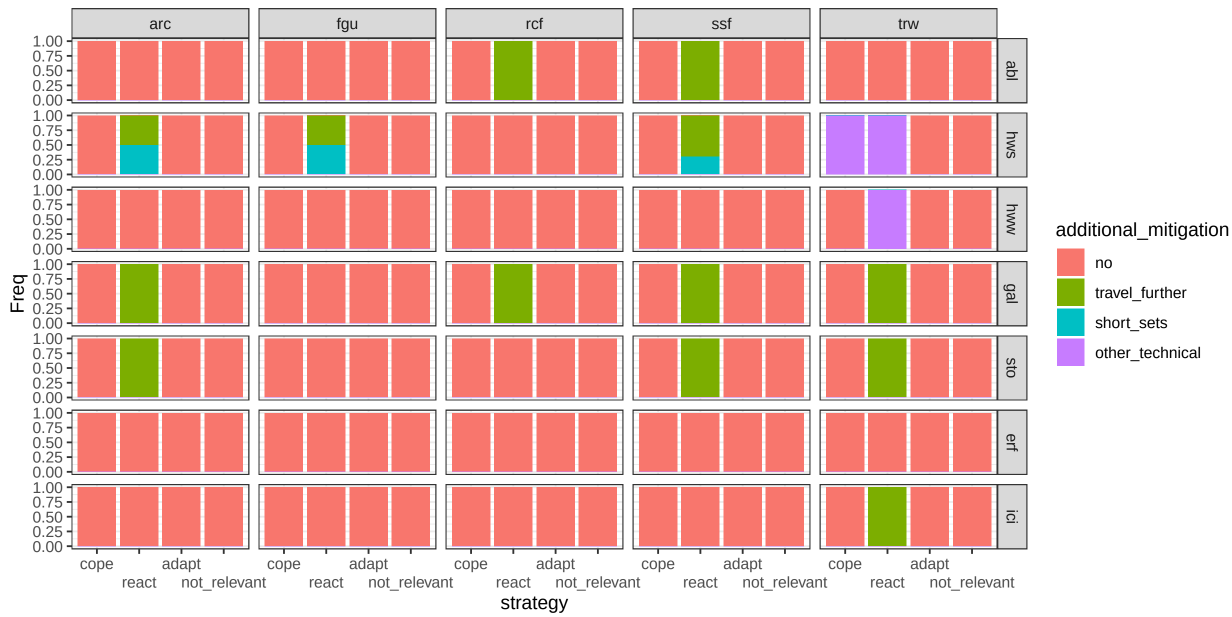

Figure B.4: conditional distribution for additional_mitigation . Columns represents levels of the node fishing_style; rows are levels of the node event.

Gear

Parents: fishing_style, event, strategy.

Levels: otm, gns_spf, gns_fws, fpn_ana, rod, icefishing, inland.

Description: Describes the most probable gear used by each fisherman. The CPT base on literature, when available. The coastal small scale fisherman and the trawlers preferences base on the FDI data of the EU data collection framework. The recreational fisher base on the Fritidsfiske 2024. In the case when the strategy is cope or react, the CPT reflects what we do expect from the fisher in normal conditions (this can be one or more gears). When the strategy is Adapt, the CPT accounts for what has been mentioned throughout the questionnaire..

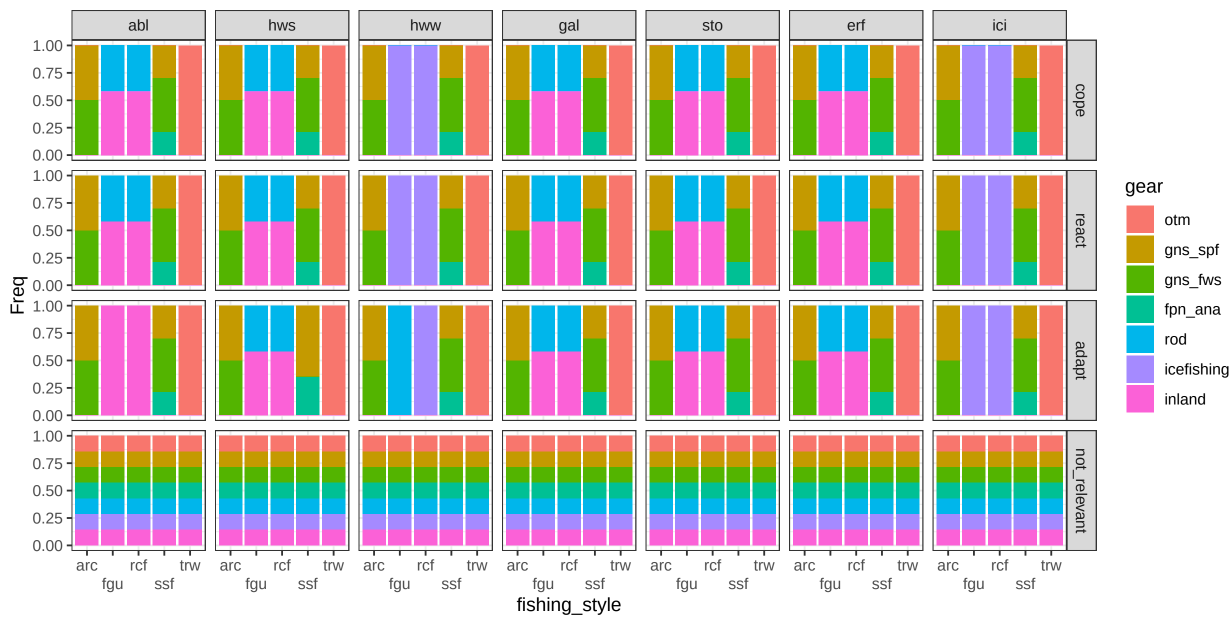

Figure B.5: conditional distribution for gear . Columns represents levels of the node event; rows are levels of the node strategy.

Catchability

Parents: target, event, gear.

Levels: worse, same, better.

Description: the ability to catch fish depending on gear-fish interactions. For example, a gillnet clogged in algae is more visible and less efficient (less catchability); a gillnet in murky waters is less visible and more efficient (more catchability); the necessity to go to haul the net sooner because of heatwave diminish the soaking time (less catchability). For each combination of species and even the CPT is filled with method M1-bis based on the answer to Q3 (catchability)..

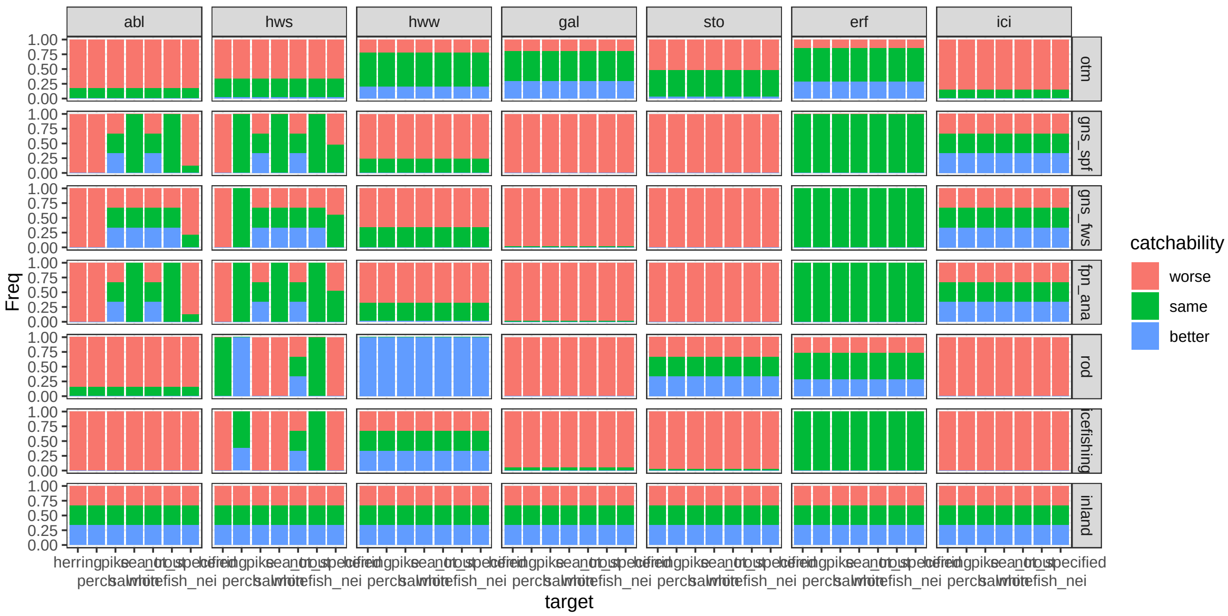

Figure B.6: conditional distribution for catchability . Columns represents levels of the node event; rows are levels of the node gear.

Catch condition

Parents: target, event, additional_mitigation.

Levels: worse, same.

Description: the real or perceived quality of catches in terms of aspect/health status. Does NOT refer to species composition or distribution. When the technical solution is no, the CPT is filled with method M1-bis based on the answer to Q5 (catch_condition). The combination of species, events and technical solutions are inspected to identify which of those works. For these , the catch condition is same with probability = 1.

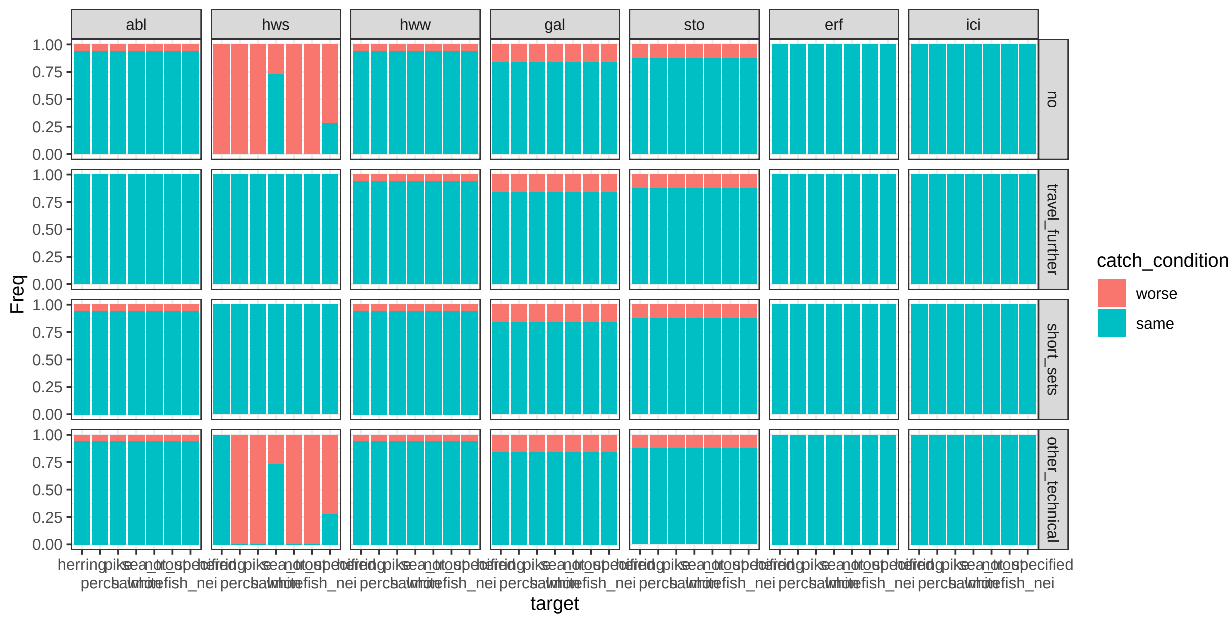

Figure B.7: conditional distribution for catch_condition . Columns represents levels of the node event; rows are levels of the node additional_mitigation.

Personal safety

Parents: event, gear, additional_mitigation.

Levels: no, minor, major.

Description: any variation to ordinary working condition that can increase the potential physical damage for operators (i.e.: icing can require more time on the deck that translates in increased risk of slipping; a violent storm is life threatening). When the technical solution is no, the CPT is filled with method M1-bis based on the answer to Q9 (damage). When the technical solution is "travel further", the personal safety is "no" with probability 1. With the available answers no other measure is implemented to deal with personal safety..

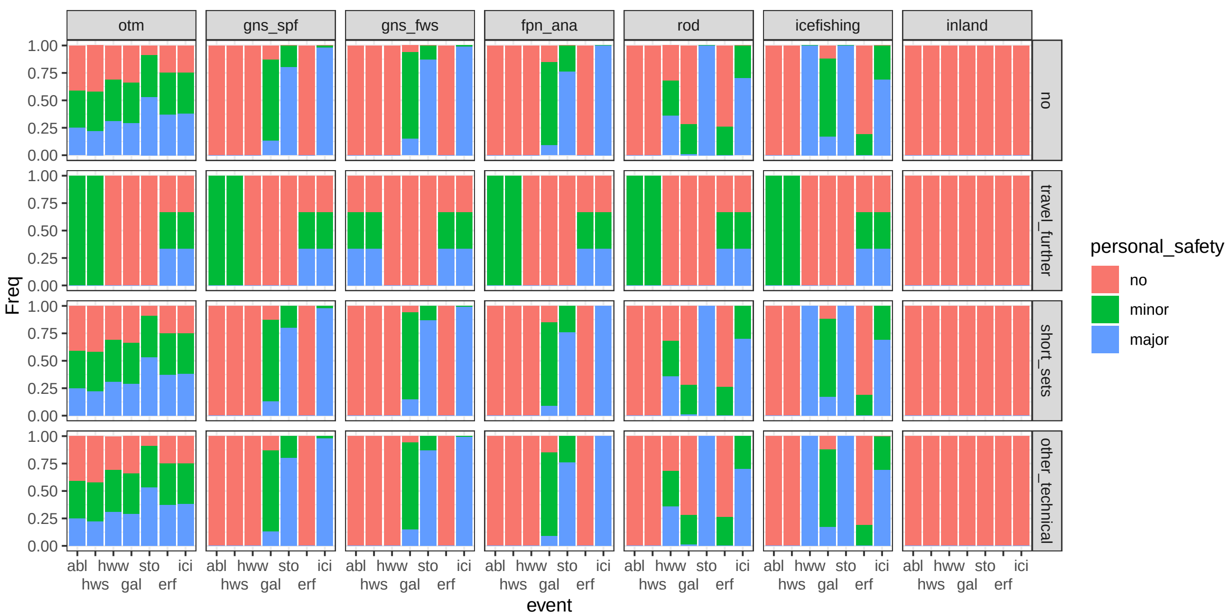

Figure B.8: conditional distribution for personal_safety . Columns represents levels of the node gear; rows are levels of the node additional_mitigation.

Damage

Parents: event, gear, additional_mitigation.

Levels: no, minor, major, destroy.

Description: damage to the fishing gear ranging from minor to completely losing the fishing gear or part of it because it was lost or irreparably damaged. When the technical solution is no, the CPT is filled with method M1-bis based on the answer to Q10 (damage). When the technical solution is "travel further", the damage is "no" with probability 1. With the available answers no other measure is implemented to deal with damages..

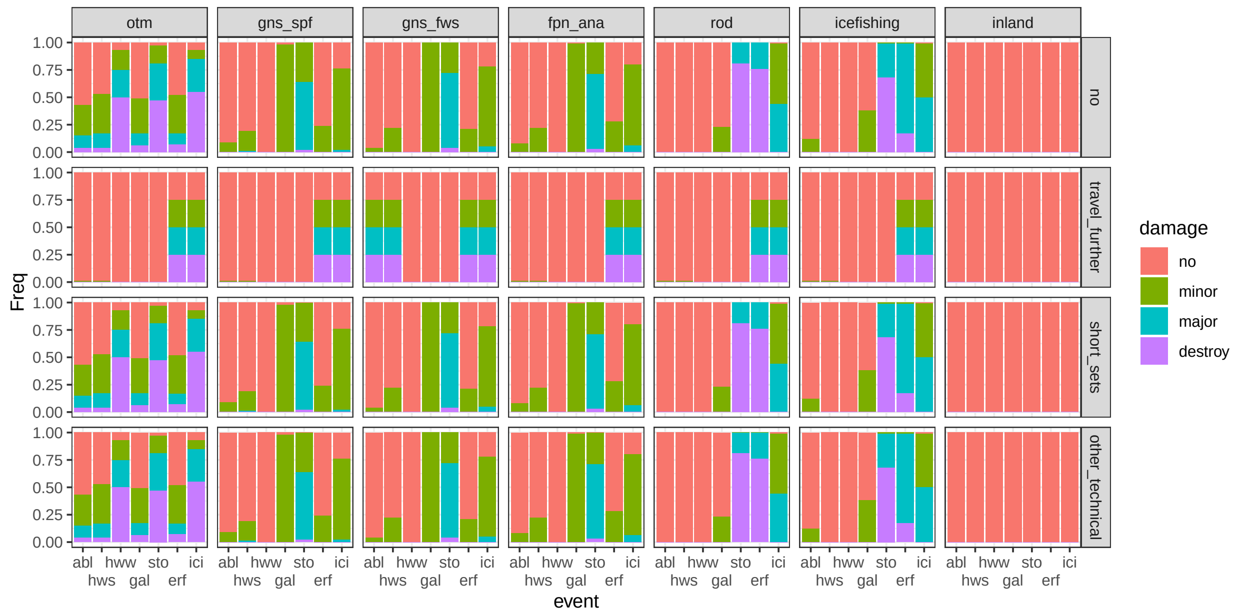

Figure B.9: conditional distribution for damage . Columns represents levels of the node gear; rows are levels of the node additional_mitigation.

Target

Parents: gear, strategy, event.

Levels: herring, perch, pike, salmon, sea_trout, whitefish_nei, not_specified.

Description: Describes the most probable target species associated to each fishing gear. The CPT base on literature, when available. The coastal small scale fisherman and the trawlers preferences base on the FDI data of the EU data collection framework. The recreational fisher base on the Fritidsfiske 2024. In the case when the strategy is cope or react, the CPT reflects what we expect from the fisher in normal conditions (this can be one or more combinations of gear/species). When the strategy is Adapt, the CPT accounts for what has been mentioned throughout the questionnaire. Example: gillnetters normally target a set of species (herring, whitefish, perch), but when there is an heatwave they switch primarily to perch..

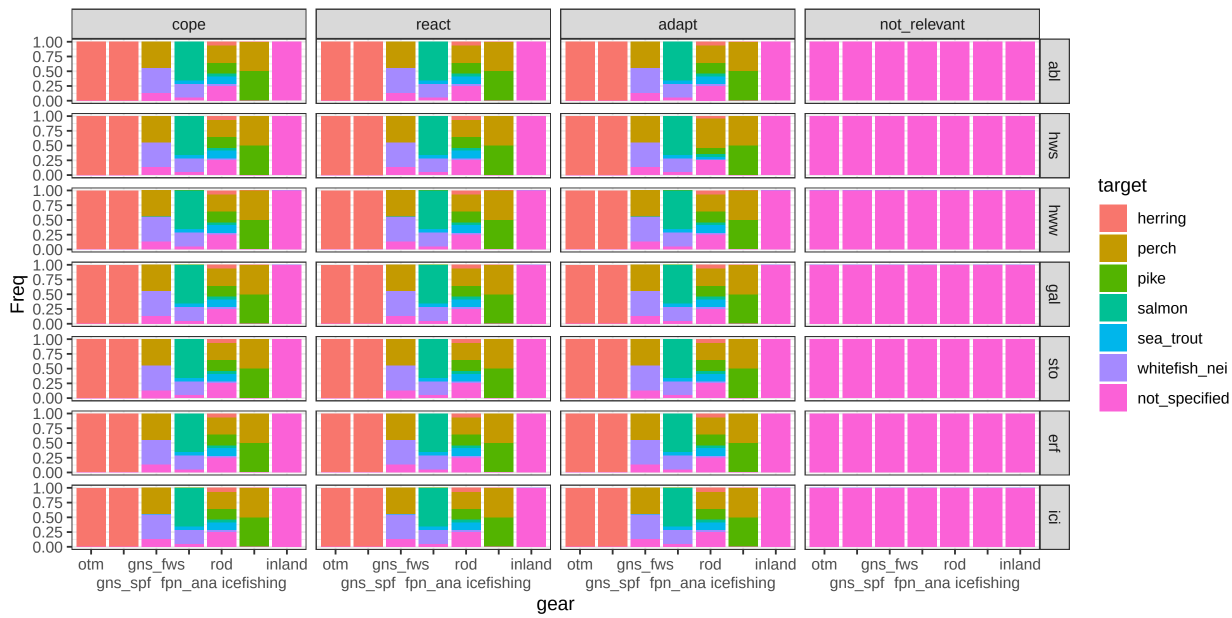

Figure B.10: conditional distribution for target . Columns represents levels of the node strategy; rows are levels of the node event.

Go out

Parents: event, fishing_style, strategy.

Levels: no, yes.

Description: Describes the probability to go out fishing. When the parent node strategy is “cope”, the CPT is filled with method M1 based on the answer to Q2 (go_out). In the case when the parent node strategy is “react” or “adapt”, the decision is always to go fishing..

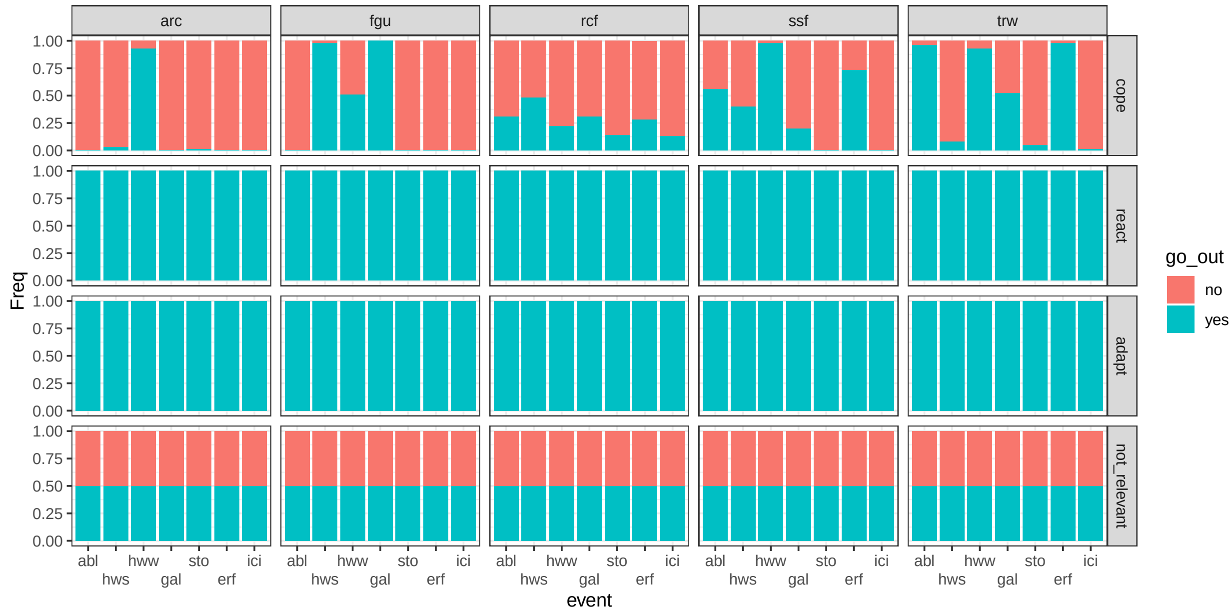

Figure B.11: conditional distribution for go_out . Columns represents levels of the node fishing_style; rows are levels of the node strategy.

Chance nodes - User

Stress

Parents: personal_safety, damage, catch_condition, fishing_style.

Levels: no, yes.

Description: psychological aspect, elements such as anxiety and sense of fatigue which can eventually drives motivation (Chatterjee and and Bolar, 2019). The stressful events considered are having damages to the gear, being injuried, being the catch in poor condition. The CPT is filled with strategy M2, where the answers to Q21 a, b and c (stress) are used as weights..

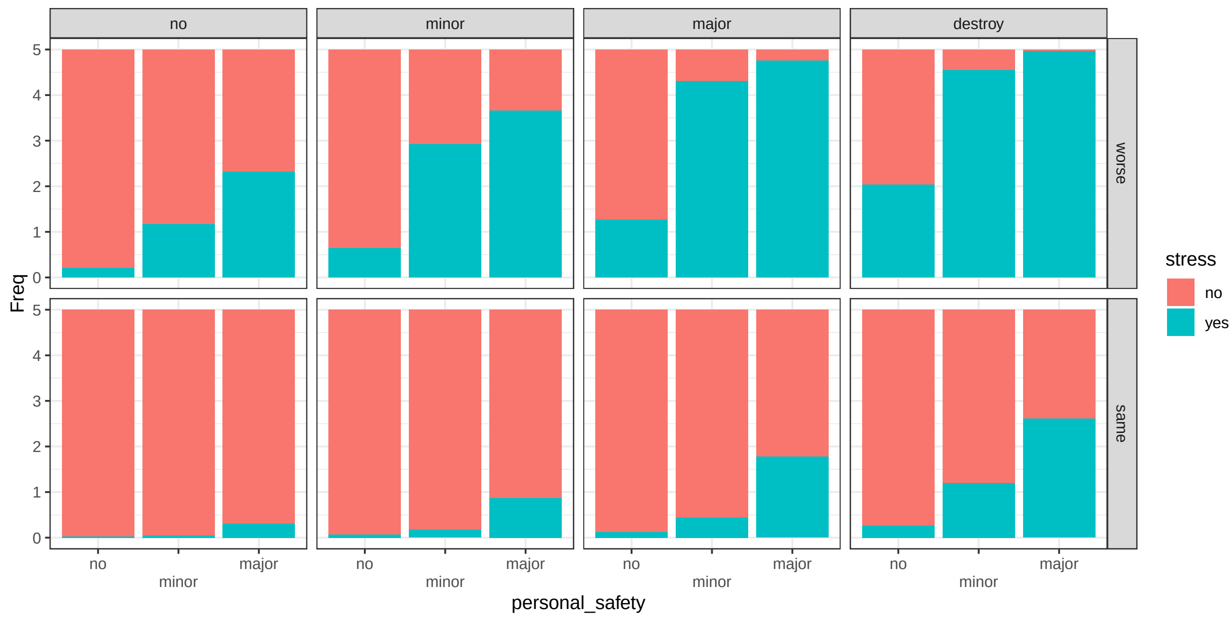

Figure B.12: conditional distribution for stress . Columns represents levels of the node damage; rows are levels of the node catch_condition.

Catches

Parents: go_out, stock_status, catchability.

Levels: no, poor, average, good.

Description: relaised catched.

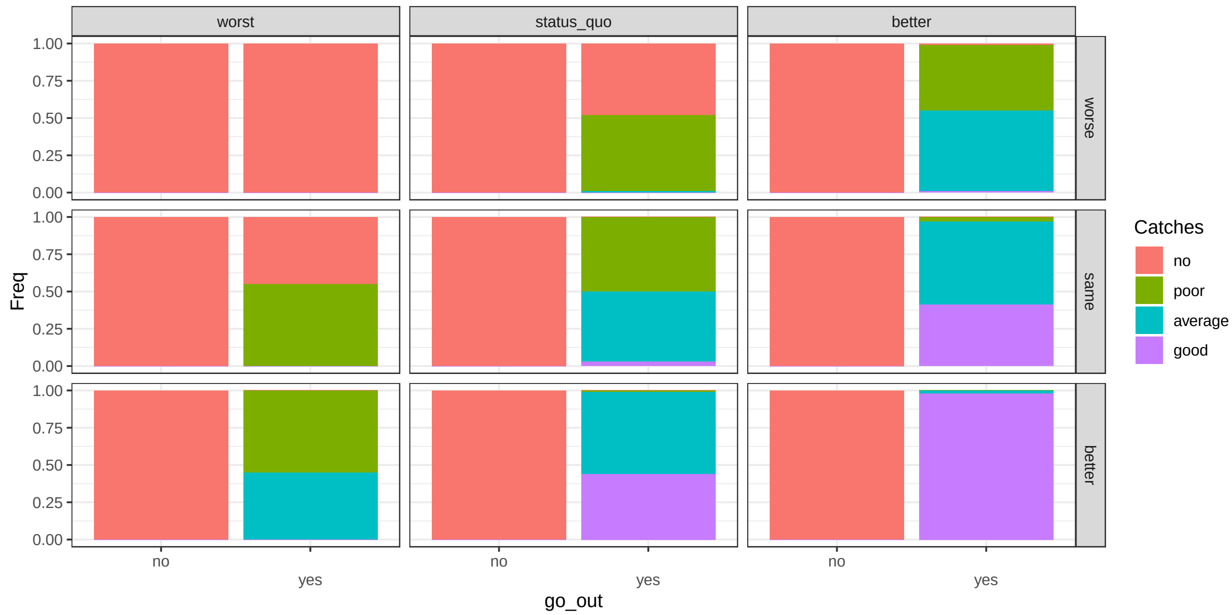

Figure B.13: conditional distribution for Catches . Columns represents levels of the node stock_status; rows are levels of the node catchability.

Health

Parents: stress, personal_safety.

Levels: Low, Medium, High.

Description: divorced node that summarizes variables related to physical and mental health. All the conditions are necessary to obtain a large value; therefore, we implement the weighted min algorithm (Eq.). Also, the contribution is uneven, with stress contributing 20% less than injury.

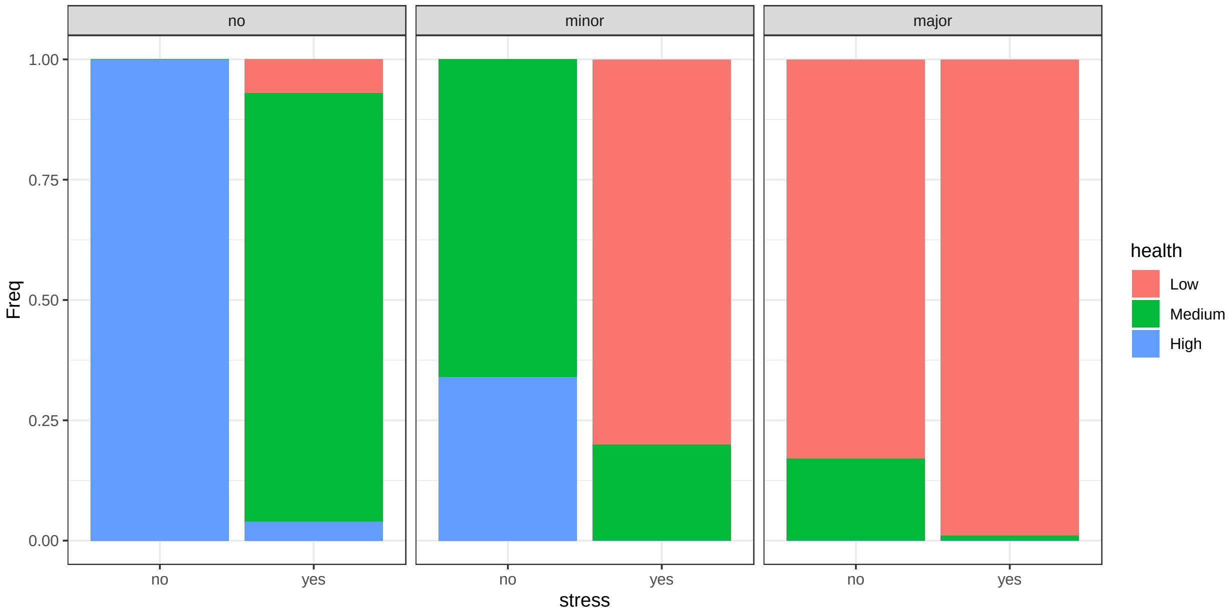

Figure B.14: conditional distribution for health . Columns represents levels of the node personal_safety.

Satisfaction

Parents: Catches, catch_condition.

Levels: Low, Medium, High.

Description: Summarizes variables related to the non-monetary satisfaction derived from the fishing experience. This includes the time spent fishing, the catch amount and the average fish size. All the variables are calculated as relative change. All the conditions are necessary to obtain a large value; therefore, we implement the weighted min algorithm (Eq.). In this case we give equal weight.

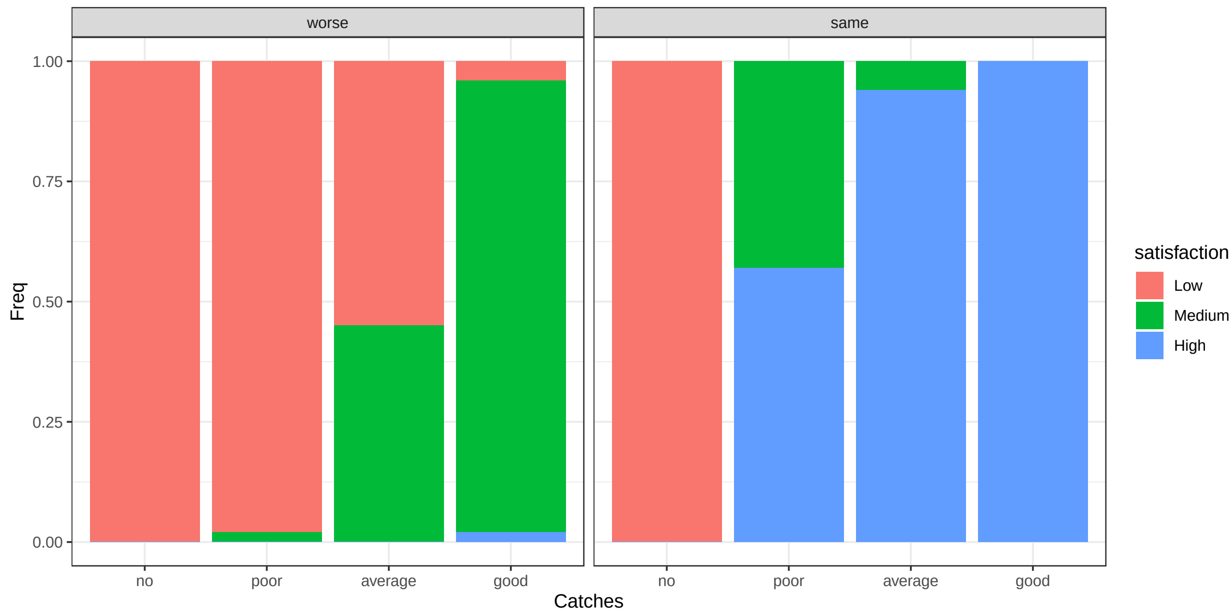

Figure B.15: conditional distribution for satisfaction . Columns represents levels of the node catch_condition.

Chance nodes - Economic

Cost

Parents: additional_mitigation, damage, fishing_style.

Levels: no, low, medium, high.

Description: Cost represent how large is the expense associated with an extreme weather event compared to the annual expected costs. No is < 0.5%; Low is between 0.5% and 2.5%; Medium is between 2.5% and 5%; high is larger than 5%. The CPT base on literature, when available. The coastal small scale fisherman and the trawlers costs base on the Annual Economic Report data of the EU data collection framework..

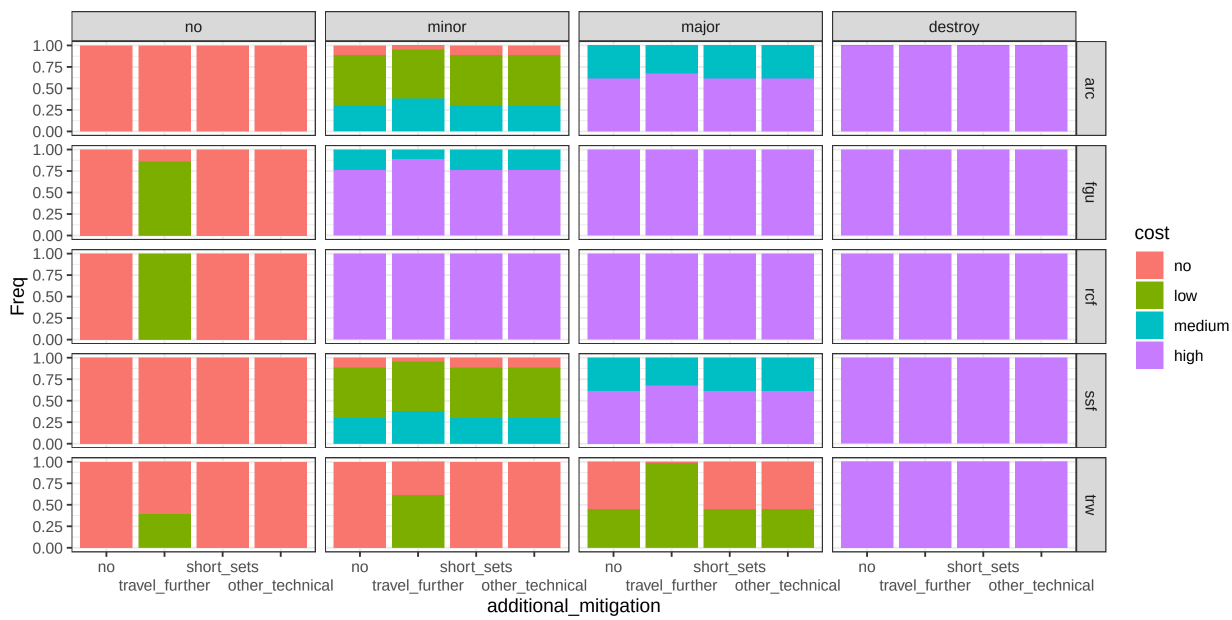

Figure B.16: conditional distribution for cost . Columns represents levels of the node damage; rows are levels of the node fishing_style.

Non monetary_value

Parents: health, satisfaction, go_out.

Levels: Low, Medium, High.

Description: summarizes the outcomes that are not related with money. It depends on the possibility to go out fishing and to obtain satisfying catches, while staying safe. In this case all the conditions are necessary to obtain a large value, therefore we implement the weighted min algorithm (Eq.). The health node is Low-Medium-High; the satisfaction is below-average-more.

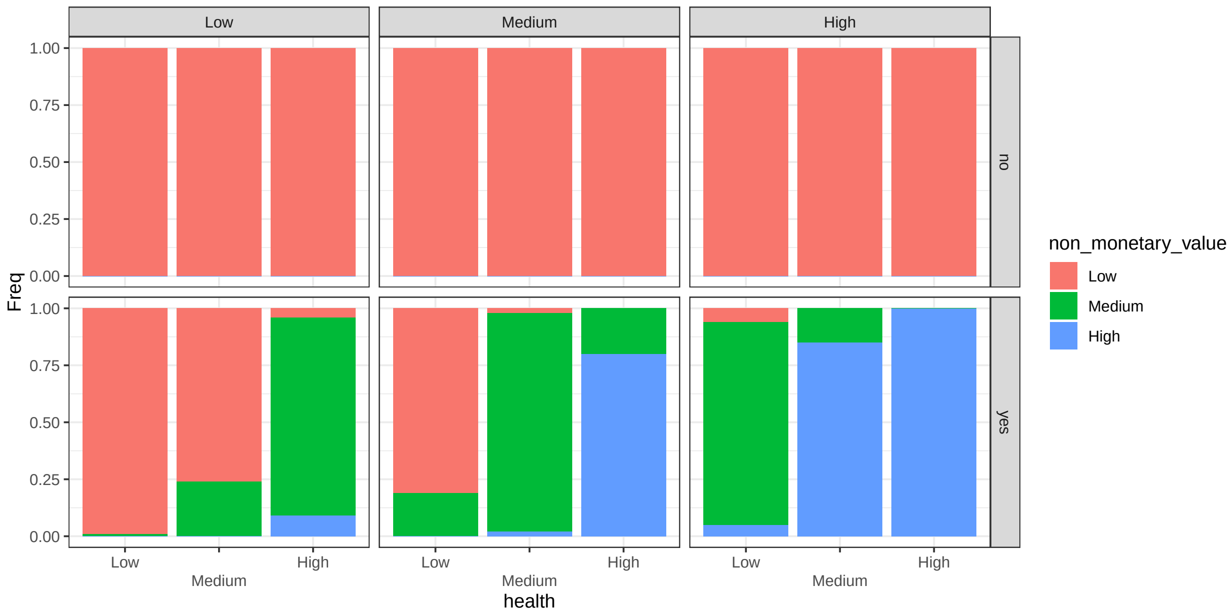

Figure B.17: conditional distribution for non_monetary_value . Columns represents levels of the node satisfaction; rows are levels of the node go_out.

Credit

Parents: fishing_style.

Levels: difficult, neutral, easy.

Description: The possibility to obtain loans for investments.

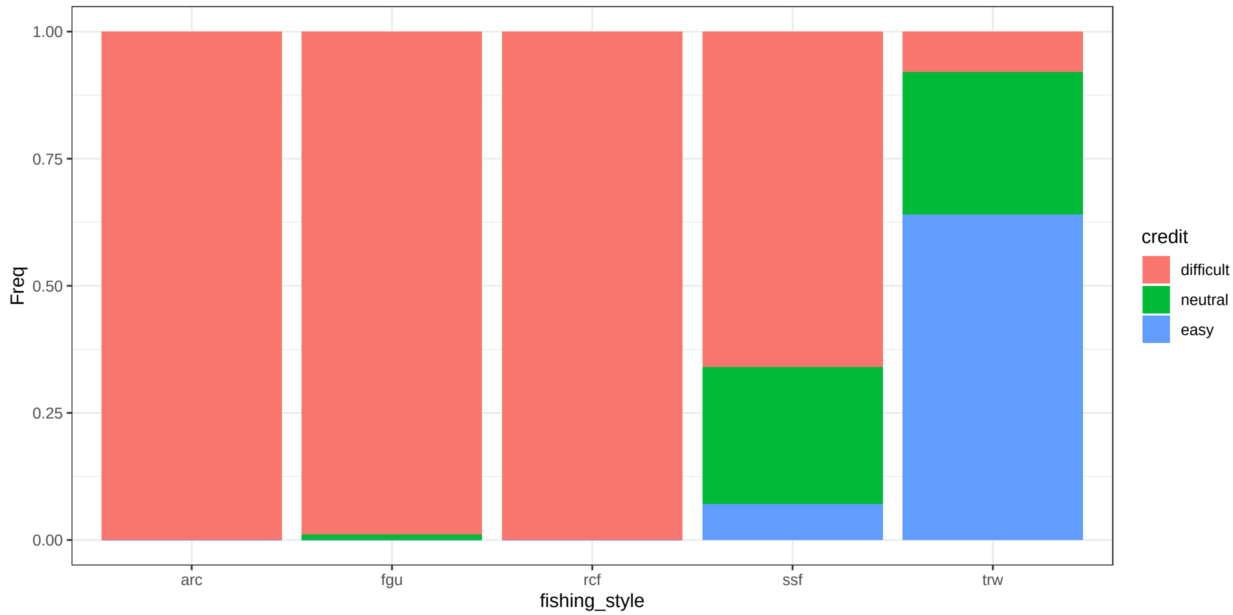

Figure B.18: conditional distribution for credit .

Entrepeneurship

Parents: fishing_style.

Levels: low, medium, high.

Description: Motivation and capabilities of individuals to explore new technologies and methods for fishing and marketing.

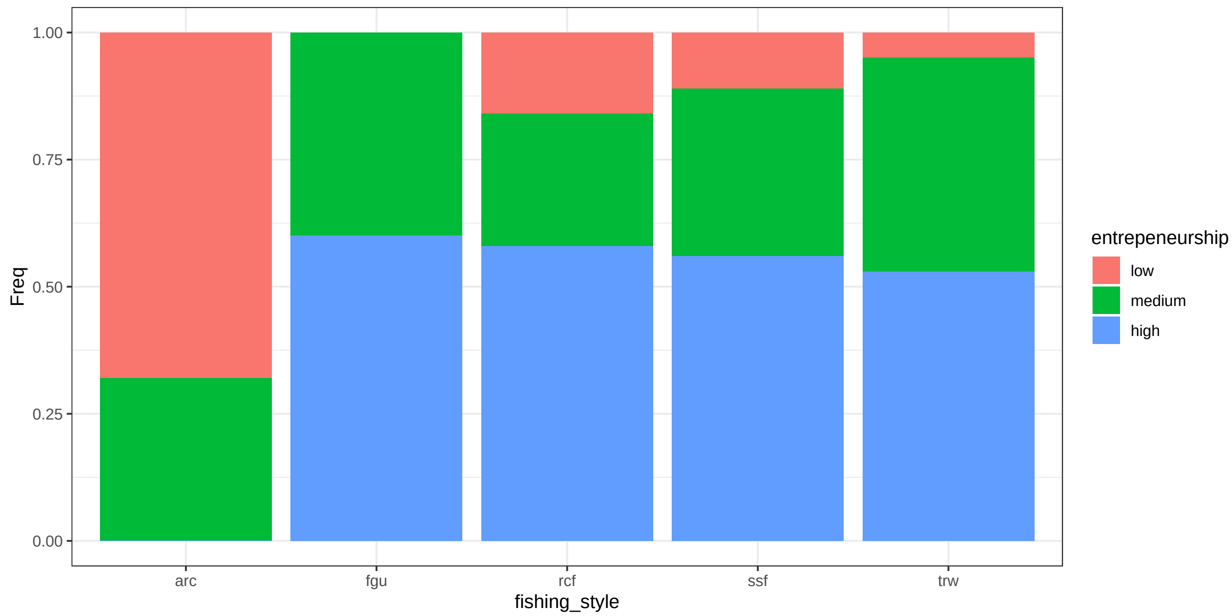

Figure B.19: conditional distribution for entrepeneurship .

Innovative capacity

Parents: credit, entrepeneurship.

Levels: low, medium, high.

Description: The capacity to pursue different avenues (in fishing, marketing, technology, etc.) and to adapt to changing circumstances.

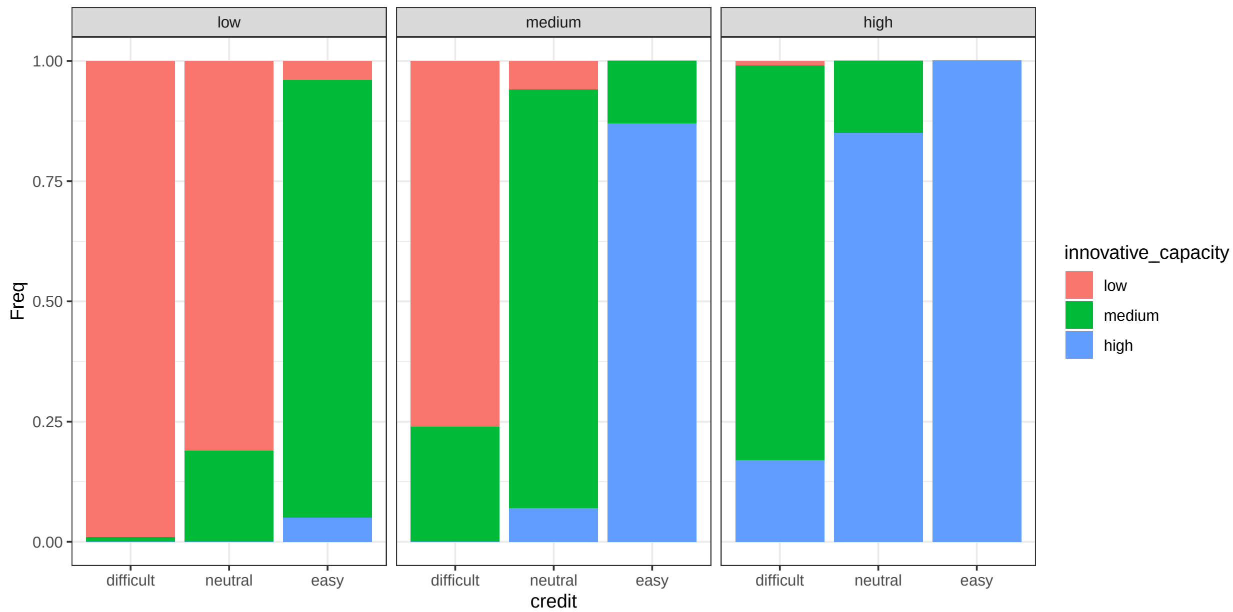

Figure B.20: conditional distribution for innovative_capacity . Columns represents levels of the node entrepeneurship.

Professional

Parents: fishing_style.

Levels: No, Yes.

Description: measures how fishing is a source of income. Full-time job; side-job; recreational (or leisure).

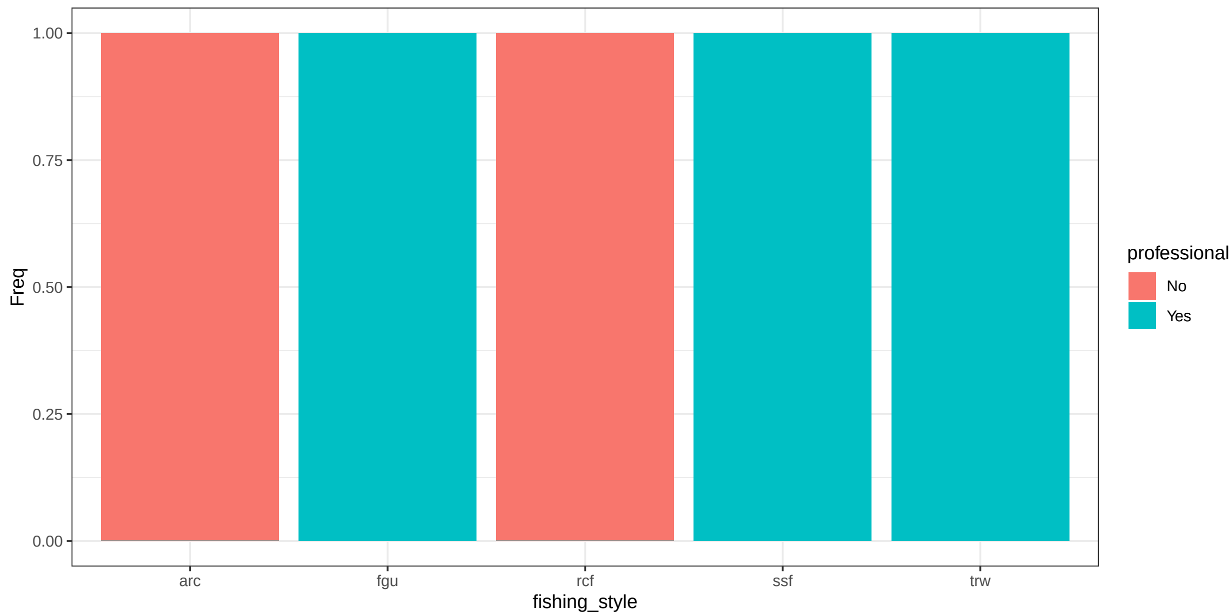

Figure B.21: conditional distribution for professional .

Job mobility

Parents: fishing_style.

Levels: low, medium, high.

Description: The extent to which non-fishing income-generating activities are accessible to an individual or household.

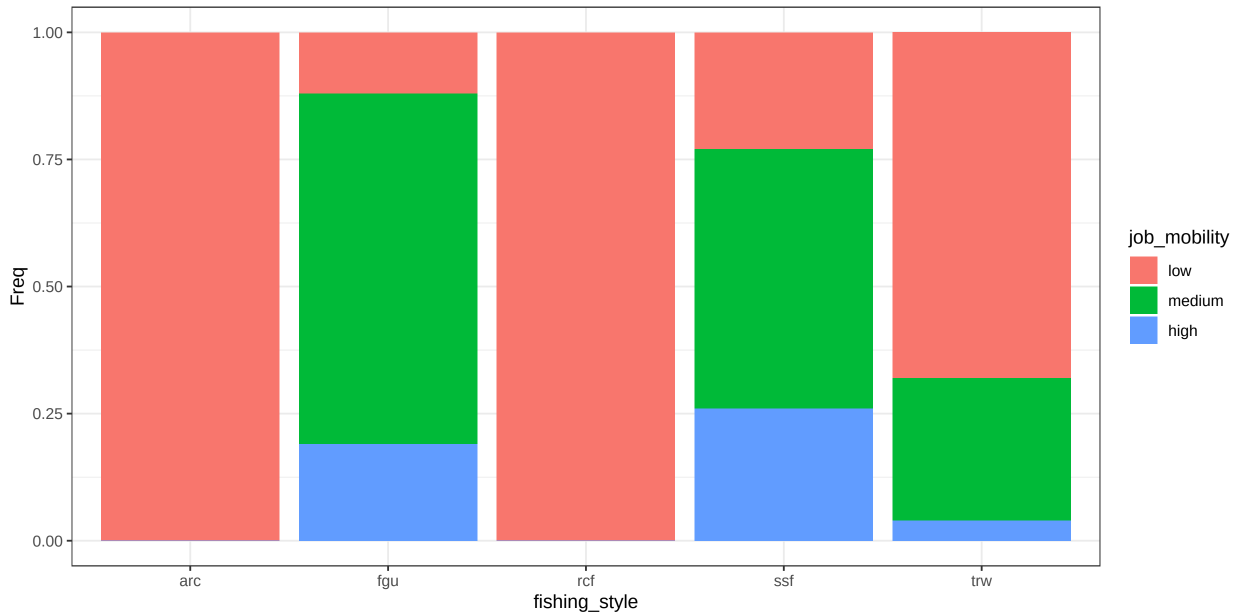

Figure B.22: conditional distribution for job_mobility .

Economic importance

Parents: job_mobility, professional, innovative_capacity.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

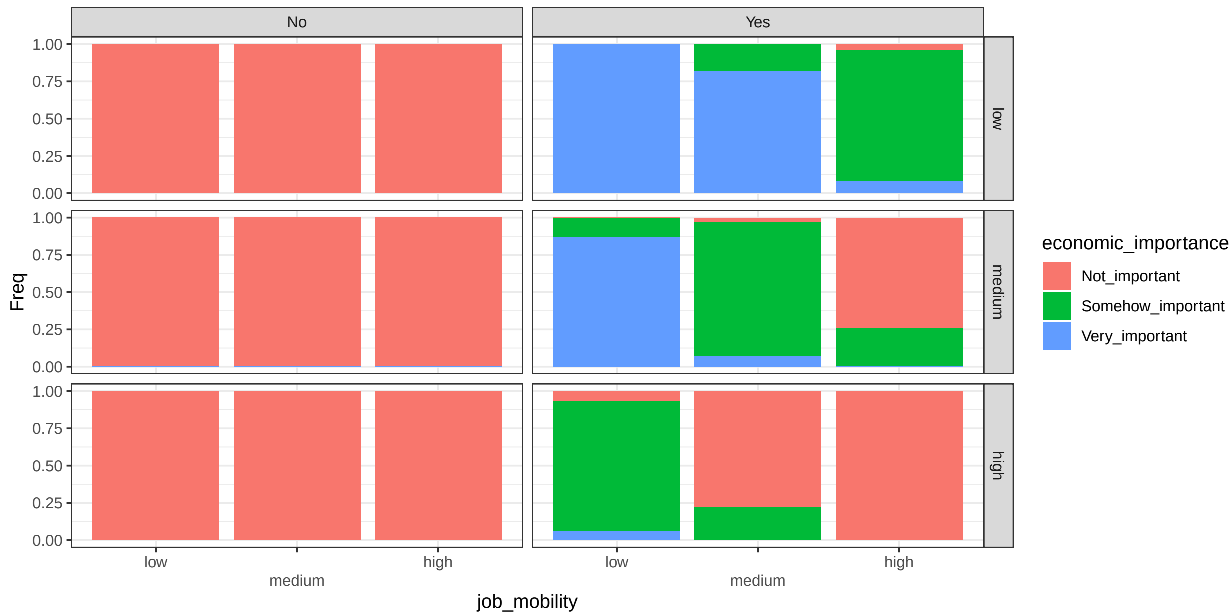

Figure B.23: conditional distribution for economic_importance . Columns represents levels of the node professional; rows are levels of the node innovative_capacity.

Chance nodes - Individual

Identity

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The degree to which fishing contributes to an individual's sense of identity, including self-perception.

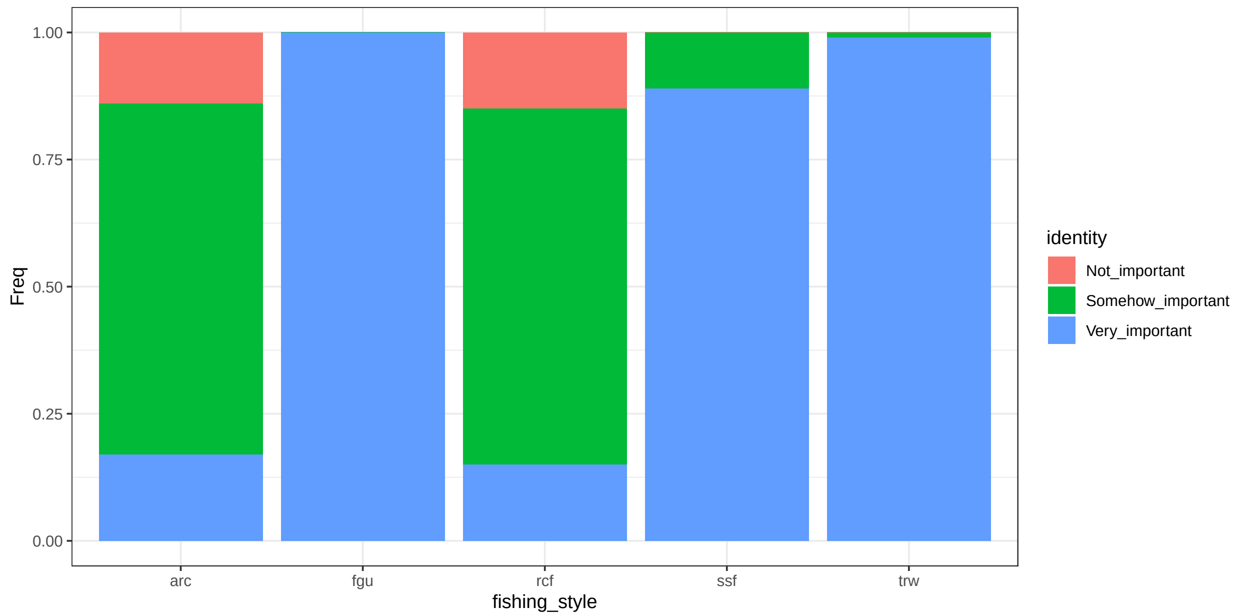

Figure B.24: conditional distribution for identity .

Home

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The degree to which fishing contributes to build a sense of place/home to a specific location. It can either be at sea or at land.

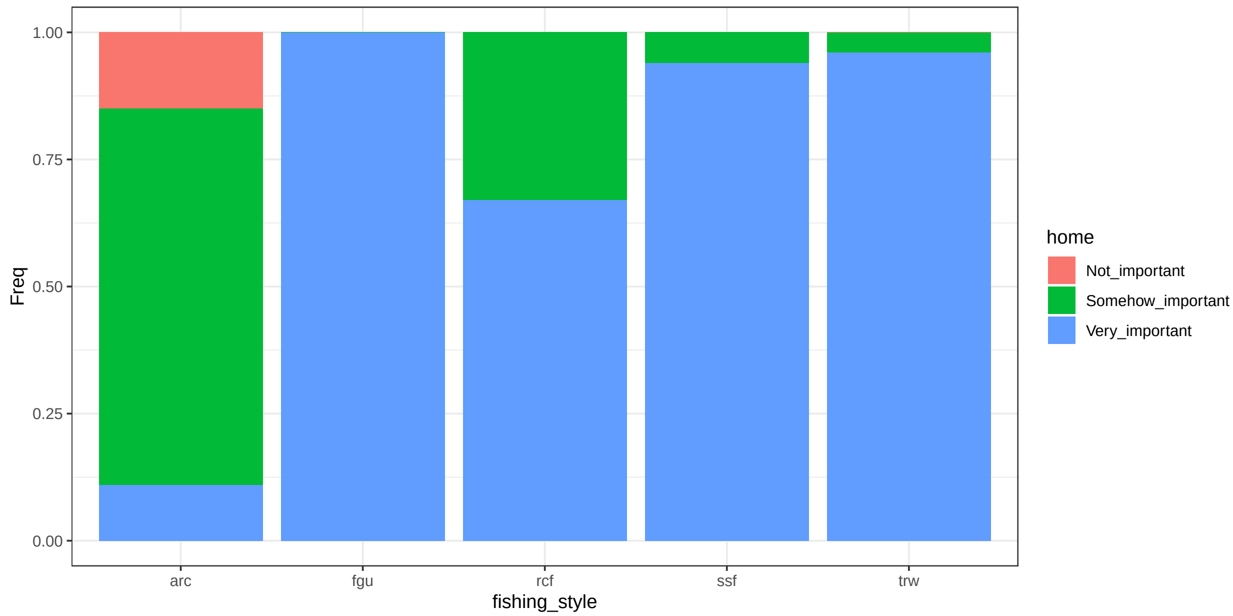

Figure B.25: conditional distribution for home .

Outdoor benefit

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The extent which an individual value the time spent outdoor (while fishing) to his personal well-being.

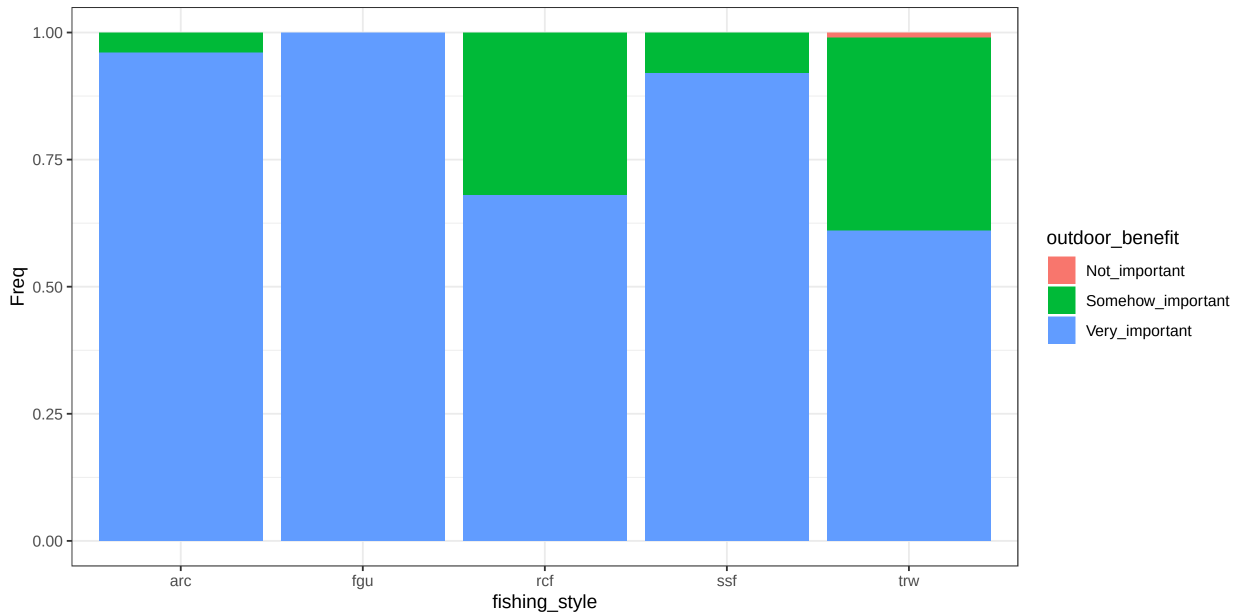

Figure B.26: conditional distribution for outdoor_benefit .

Individual importance

Parents: outdoor_benefit, identity, home.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

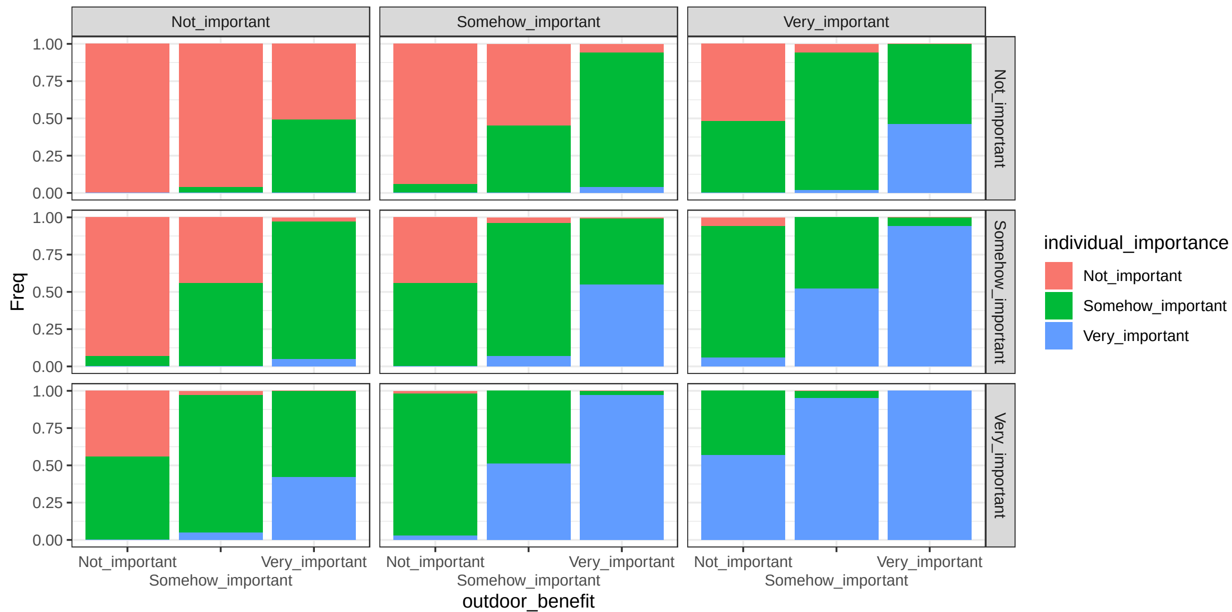

Figure B.27: conditional distribution for individual_importance . Columns represents levels of the node identity; rows are levels of the node home.

Chance nodes - Societal

Knowledge transmission

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important do you consider transmitting your fishing knowledge to future generations.

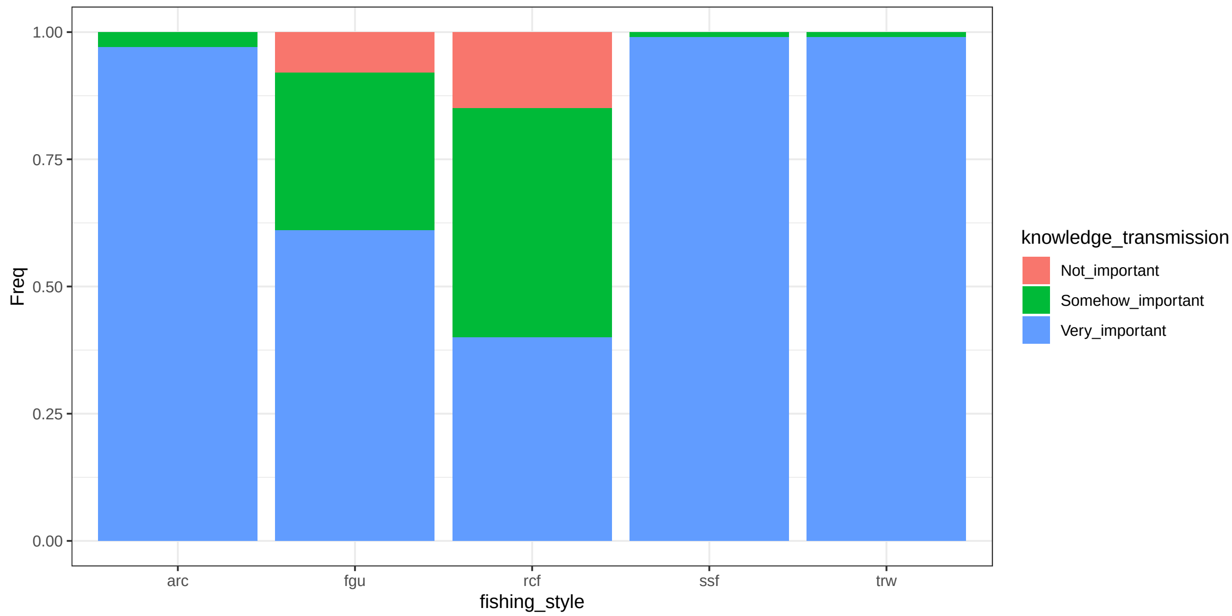

Figure B.28: conditional distribution for knowledge_transmission .

Benefit local_community

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important is fishing to maintain social relations within your community.

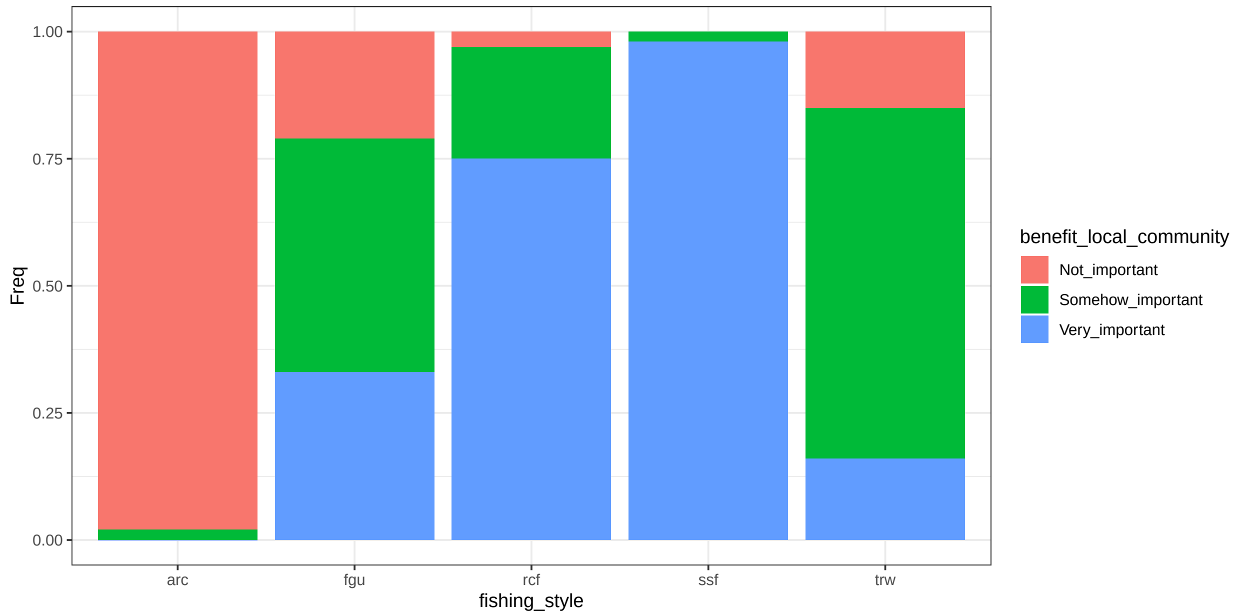

Figure B.29: conditional distribution for benefit_local_community .

Traditional food

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: is your fishing activity contributing to provide for traditional food?.

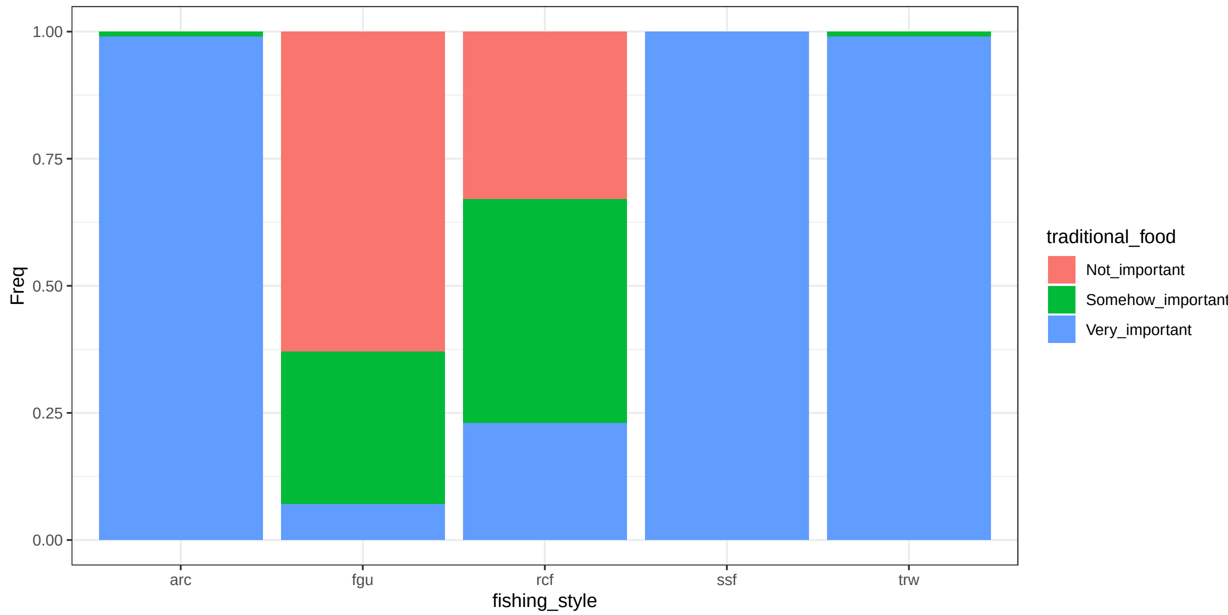

Figure B.30: conditional distribution for traditional_food .

Benefit personal_community

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important is fishing to maintain social relations within your community.

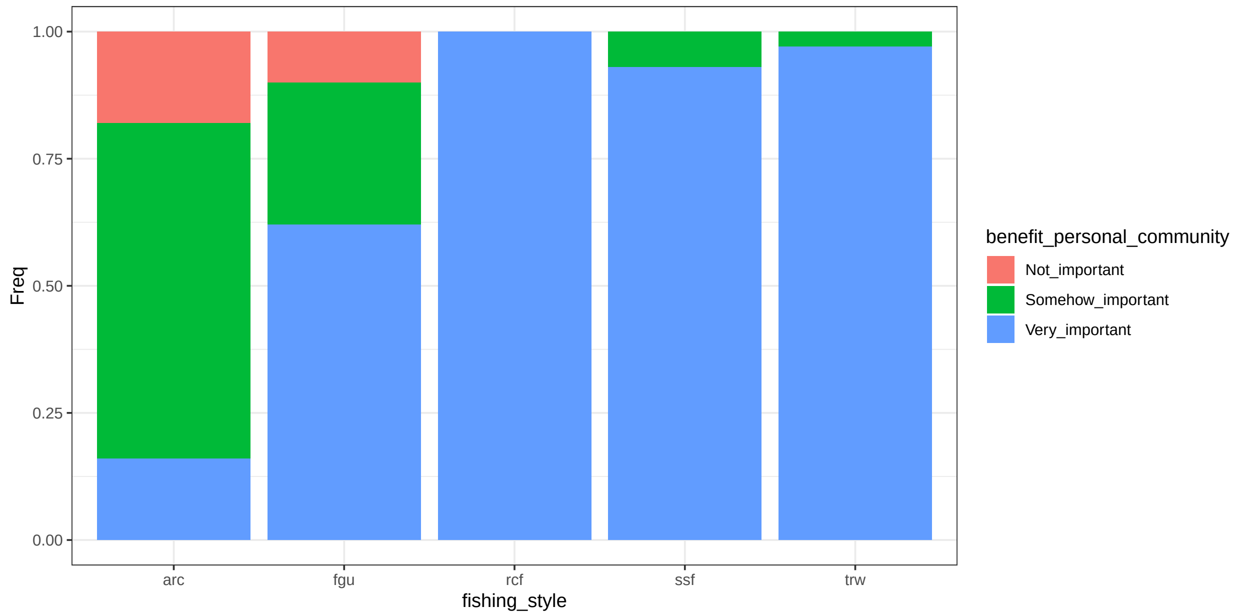

Figure B.31: conditional distribution for benefit_personal_community .

Societal importance

Parents: knowledge_transmission, benefit_local_community, traditional_food, benefit_personal_community.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

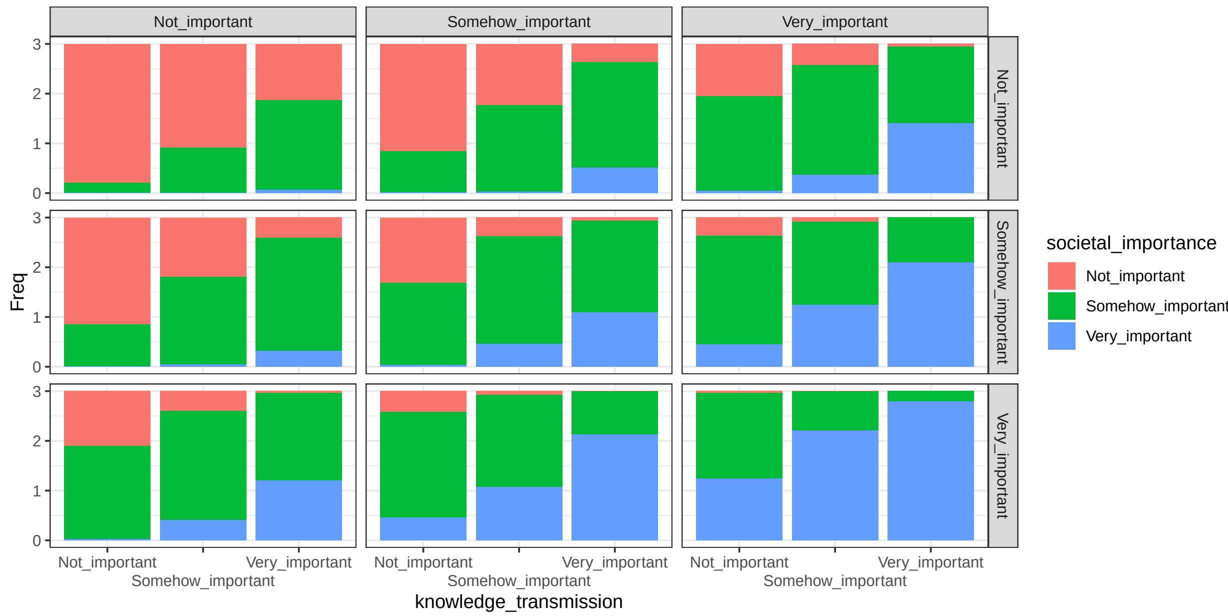

Figure B.32: conditional distribution for societal_importance . Columns represents levels of the node benefit_local_community; rows are levels of the node traditional_food.

Utilities - Risk

Shape how different fisheries will perceive risk. To each of these variables, the effect of the human community variables is to be interpreted as "how much this variable is important"? The CPT table combine the concepts of ranked nodes and the classic definition of Risk = Hazard * Vulnerability. In the case of individual and societal importance, Risk is exactly calculated as Hazard * Vulnerability. Parent categories (Low, medium and high; as well as not importand, somehow important and very important) are mapped to 0, 0.5 and 1 values. The parent value are multiplied.

Individual risk

Parents: individual_importance, non_monetary_value, strategy.

Levels: Low, Medium, High, not_relevant.

Description: aspects related to the individual well-being and satisfaction.

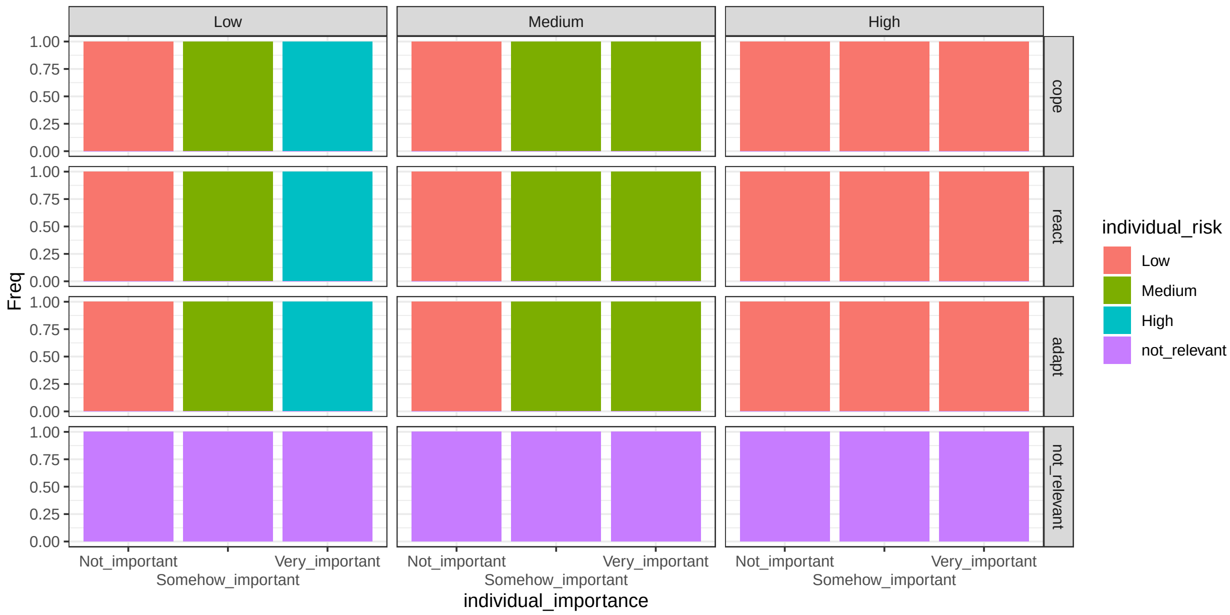

Figure B.33: conditional distribution for individual_risk . Columns represents levels of the node non_monetary_value; rows are levels of the node strategy.

Societal risk

Parents: societal_importance, non_monetary_value, strategy.

Levels: Low, Medium, High, not_relevant.

Description: aspects related to the well-being of communities.

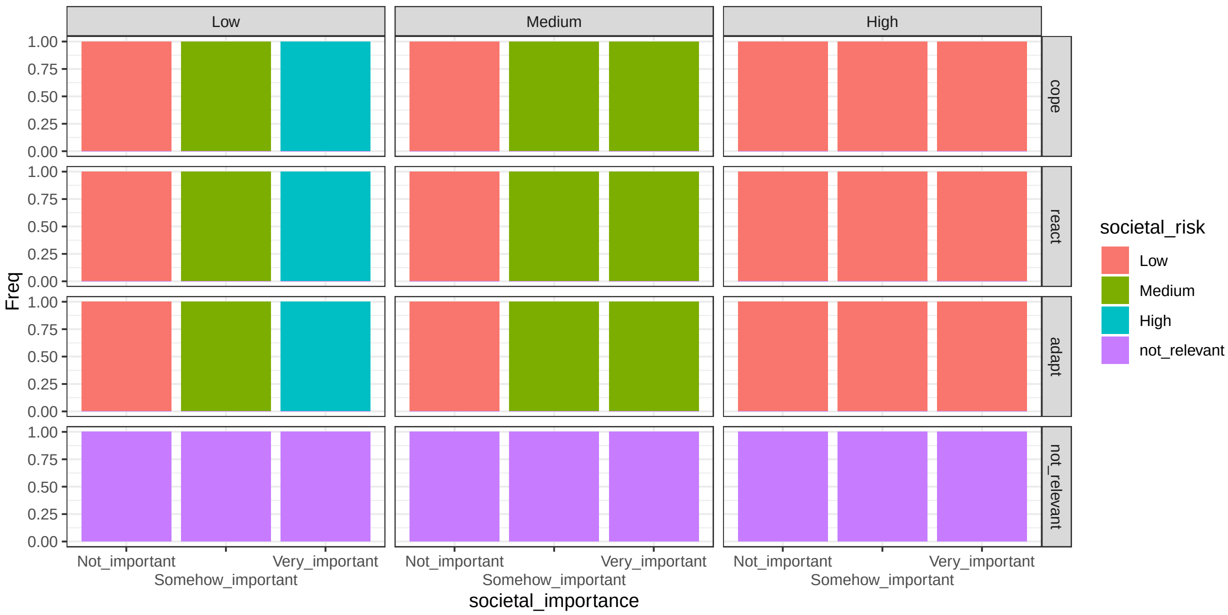

Figure B.34: conditional distribution for societal_risk . Columns represents levels of the node non_monetary_value; rows are levels of the node strategy.

Economic risk

Parents: economic_importance, cost, strategy.

Levels: Low, Medium, High, not_relevant.

Description: economic aspects related to fishing practices.

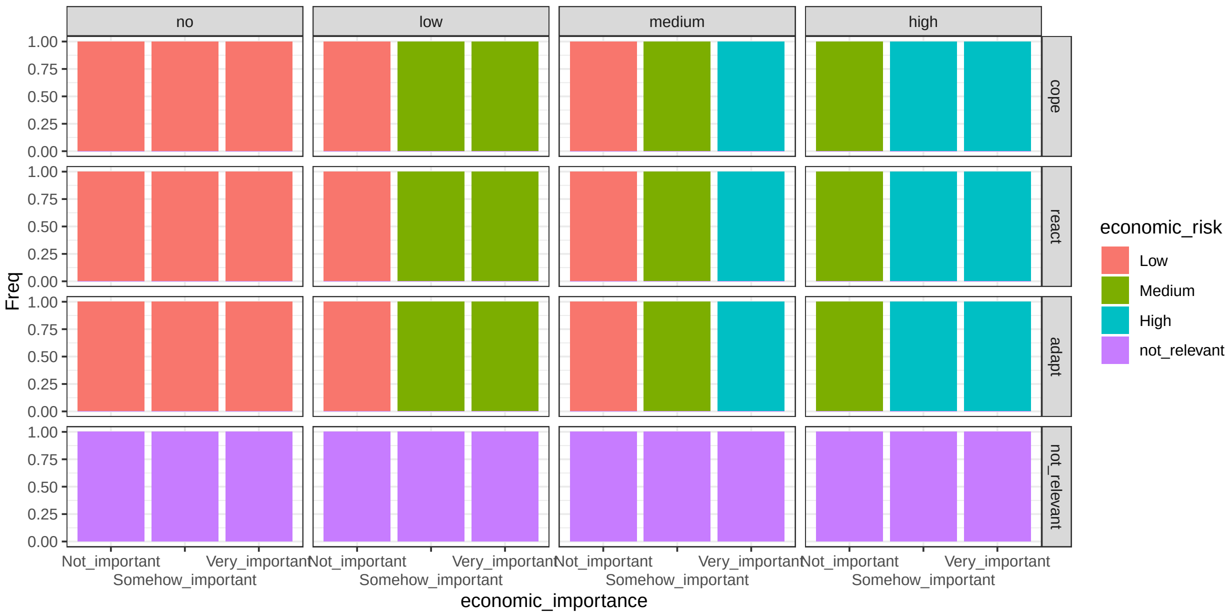

Figure B.35: conditional distribution for economic_risk . Columns represents levels of the node cost; rows are levels of the node strategy.